Algebra Essentials

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Systems of two equations

Solve the system: $x + y = 10$, $x - y = 4$.

2 [Foundational] • Target: Substitution into formulas

The formula for simple interest is $I = Prt$. Find I when $P = 2{,}000$, $r = 0.04$, $t = 3$.

3 [Foundational] • Target: Substitution into formulas

The perimeter of a rectangle is $P = 2(l + w)$. Find P when $l = 15$ and $w = 9$.

4 [Foundational] • Target: Substitution into formulas

If $v = u + at$, find v when $u = 5$, $a = 3$, $t = 4$.

5 [Foundational] • Target: Substitution into formulas

The area of a triangle is $A = \frac{1}{2}bh$. Find A when $b = 14$ and $h = 9$.

6 [Foundational] • Target: Translating words into equations

Twice a number, plus 7, equals 25. Find the number.

7 [Foundational] • Target: Translating words into equations

A number decreased by 8 is equal to three times the number, decreased by 20. Find the number.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Translating words into equations

The sum of three consecutive integers is 84. Find the integers.

9 [Intermediate] • Target: Translating words into equations

Five more than three times a number is the same as 10 less than five times the number. Find the number.

10 [Intermediate] • Target: Word problems

A rectangle has length $(x + 3)$ cm and width x cm. This question has two parts.

a If the area is 40 cm², write an equation in x.

b Solve for x.

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11 [Intermediate] • Target: Word problems

A father is 3 times as old as his son. In 12 years, he will be twice as old as his son. Find their current ages.

12 [Intermediate] • Target: Word problems

Tickets cost 25 riyals for adults and 15 riyals for children. A group of 10 people paid 220 riyals in total. How many adults were in the group?

13 [Intermediate] • Target: Word problems

The sum of two numbers is 45, and their difference is 11. Find the two numbers.

14 [Intermediate] • Target: MCQ — Expanding and simplifying

Solve the linear equation for x: $4(2x - 3) - 3(x + 1) = 14$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: MCQ — Factoring • Cross-Lesson: 7.01 Data Analysis & Probability

In a survey of 200 students regarding their book reading habits, the data was recorded in a two-way probability table. The following table shows the results of the survey: | | Novels | Science | History | Total | |---------|--------|---------|---------|-------| | Boys | 55 | 35 | 10 | 100 | | Girls | 45 | 40 | 15 | 100 | | Total | 100 | 75 | 25 | 200 | If a student is chosen at random from these students, what is the probability that the student is a girl and reads science books?

16 [Challenging] • Target: MCQ — Solving quadratic equations • Cross-Lesson: 8.01 Word Problems & Reasoning

If $\frac{1}{x} + \frac{1}{y} = \frac{1}{3}$ and $x + y = 8$, find the value of the product xy.

17 [Challenging] • Target: MCQ — Systems of two equations • Cross-Lesson: 1.01 Arithmetic & Number Sense

Pipe A fills a swimming pool in 4 hours, and Pipe B fills it in 6 hours. If both pipes are opened simultaneously, in how many hours will the pool be full?

18 [Challenging] • Target: MCQ — Laws of exponents • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

A train travels from Riyadh to Dammam at 100 km/h and returns at 150 km/h. What is the average speed of the train for the complete round trip?

19 [Challenging] • Target: MCQ — Inequalities • Cross-Lesson: 4.01 Geometry & Measurement

A father's age is three times his son's age. If the sum of their ages is 48 years, how old is the son?

20 [Challenging] • Target: MCQ — Reading function graphs • Cross-Lesson: 5.01 Quantitative Comparisons

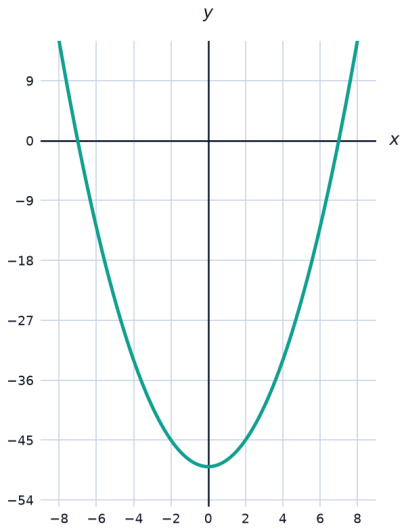
A car travels 240 km in 3 hours. At this same constant speed, how many kilometers will it travel in 5 hours?

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 **[Foundational]** • **Target: MCQ — Solving quadratic equations**

The graph shows $y = x^2 - 49$. What are its roots (where the curve crosses the x-axis)?



a. x = 7 only

b. x = 49

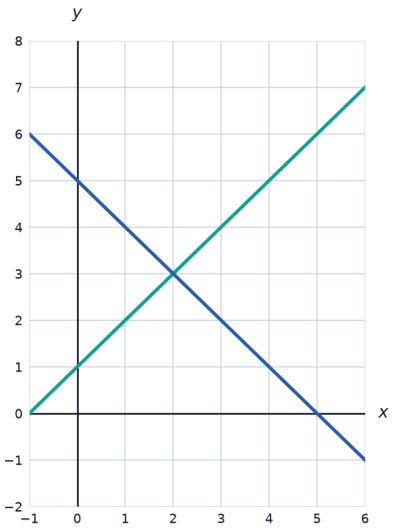
c. x = 7 or x = -7

d. x = -7 only

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22 [Foundational] • Target: MCQ — Systems of two equations

The graph shows two lines. What is their point of intersection?



a. (1, 4)

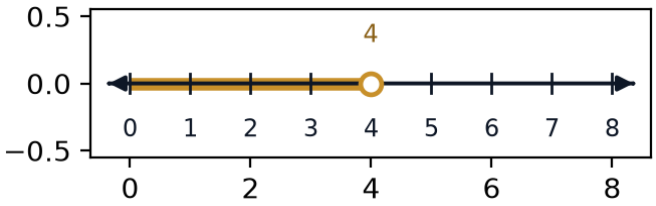
b. (0, 5)

c. (2, 3)

d. (3, 2)

23 [Foundational] • Target: MCQ — Inequalities

The number line shows the solution set of an inequality. Which inequality matches?



a. $x \geq 4$

b. $x \leq 4$

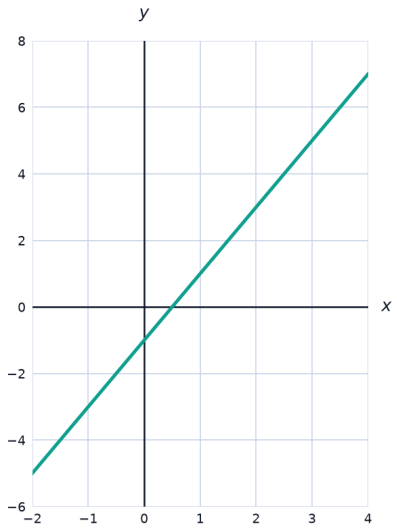
c. $x > 4$

d. $x < 4$

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24 [Foundational] • Target: MCQ — Reading function graphs

The graph shows a line. What is its slope?



a. $\frac{1}{2}$

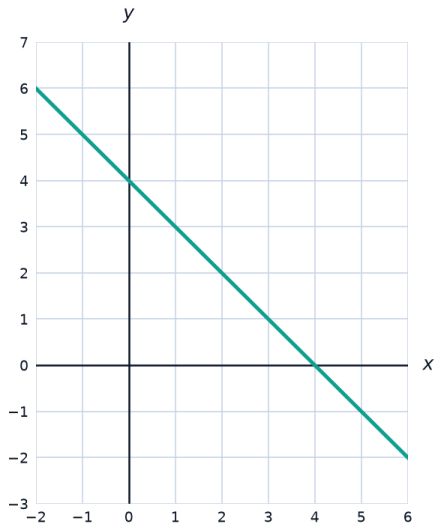
b. 2

c. -2

d. -1

25 [Foundational] • Target: MCQ — Reading function graphs

What is the y-intercept of the line shown?



a. 4

b. 0

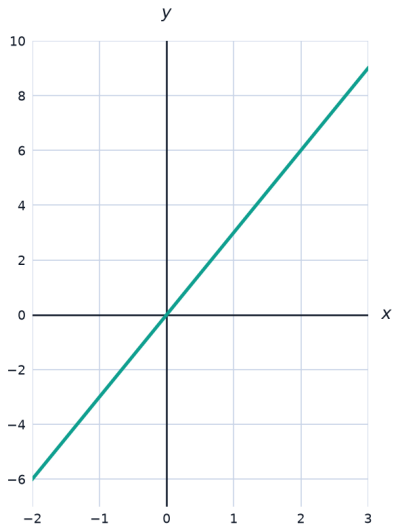
c. -4

d. 1

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26 [Foundational] • Target: MCQ — Reading function graphs

What is the equation of the line shown?



a. $y = 3x + 1$

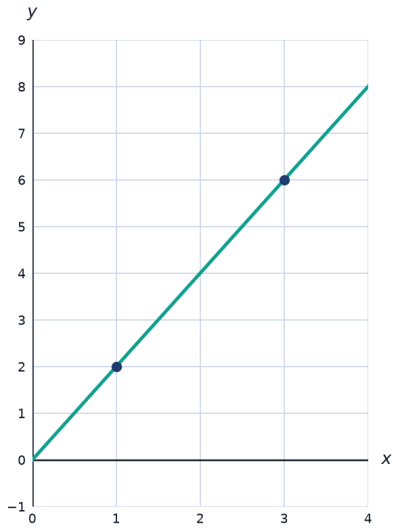
b. $y = x + 3$

c. $y = \frac{1}{3}x$

d. $y = 3x$

27 [Foundational] • Target: MCQ — Reading function graphs

Find the slope of the line through the two marked points.



a. 2

b. $\frac{1}{2}$

c. -2

d. 4

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28 [Foundational] • Target: MCQ — Substitution into formulas

Given $A = \frac{1}{2}bh$, find A when $b = 10$ and $h = 6$.

a. 16

b. 8

c. 60

d. 30

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Substitution into formulas

The rectangle shown has area $A = lw = 48$ and length $l = 8$. Find the width w.

a. 8

b. 6

c. 384

d. 40

30 [Intermediate] • Target: MCQ — Translating words into equations

Three times a number, minus 5, equals twice the number plus 7. Find the number.

a. 19

b. 2

c. -12

d. 12

31 [Intermediate] • Target: MCQ — Translating words into equations

The sum of two consecutive even integers is 54. Find the larger integer.

a. 28

b. 27

c. 29

d. 26

32 [Intermediate] • Target: MCQ — Translating words into equations

Five less than three times a number is 25. Find the number.

a. 10

b. 20

c. 5

d. 15

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33 [Intermediate] • Target: MCQ — Translating words into equations

The perimeter of a rectangle is 54 cm. The length is 3 cm more than the width. Find the width.

a. 24

b. 12

c. 15

d. 9

34 [Intermediate] • Target: MCQ — Word problems

A number increased by 20% equals 72. Find the number.

a. 90

b. 52

c. 60

d. 86.4

35 [Intermediate] • Target: MCQ — Word problems

Ahmed is twice as old as his sister. In 5 years, the sum of their ages will be 40. Find Ahmed's current age.

a. 20

b. 10

c. 15

d. 25

36 [Intermediate] • Target: MCQ — Word problems

A positive number plus its square is 30. Find the number.

a. 30

b. -6

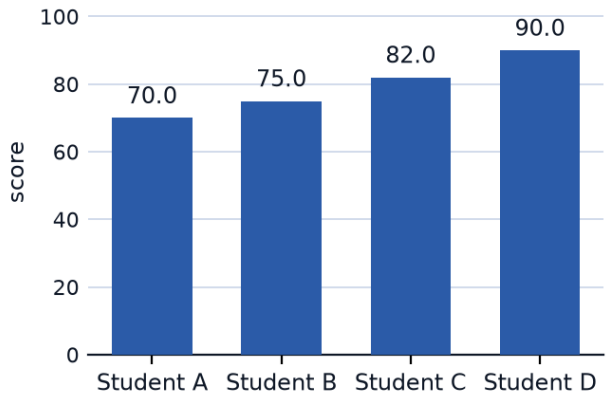
c. 6

d. 5

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37 [Intermediate] • Target: MCQ — Word problems

The bar chart shows four students' test scores. Whose score is exactly 12 more than 70?



a. Student B

b. Student D

c. Student C

d. Student A

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Word problems • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

The product of two consecutive integers is 132. Find the larger integer.

a. 13

b. 12

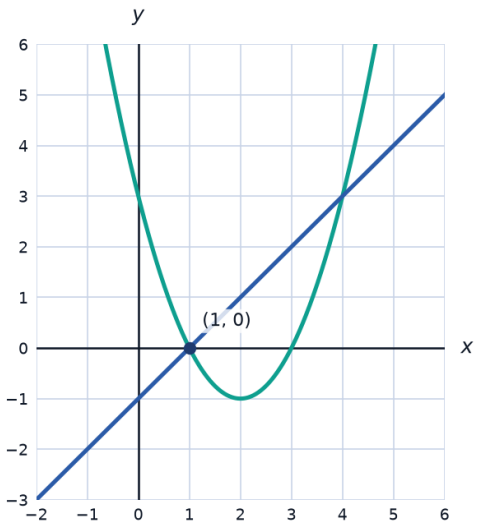
c. 10

d. 11

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39 [Challenging] • Target: MCQ — Inequalities • Cross-Lesson: 4.01 Geometry & Measurement

■■■ ■■■ $f(x) = x^2 - 4x + 3$ ■ $g(x) = x - 1$■ ■■■■■■ ■■■■■ ■■■■■■ ■■■■■■ ■■■■■■■■ ■■■■■■■■■ $f(x)$ ■ $g(x)$ ■■■■■ $x = 1$. If $f(x) = x^2 - 4x + 3$ and $g(x) = x - 1$, find the point of intersection of the graphs of $f(x)$ and $g(x)$ when $x = 1$.



a. (1, 0)

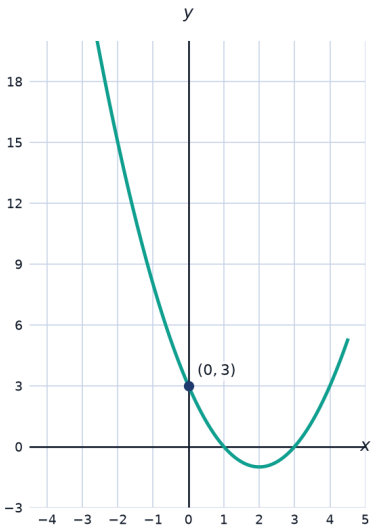
b. (0, 1)

c. (1, 1)

d. (0, 0)

40 [Challenging] • Target: MCQ — Reading function graphs • Cross-Lesson: 5.01 Quantitative Comparisons

In the given function graph, what are the coordinates of the y-intercept of the parabola $y = x^2 - 4x + 3$?



a. (0, 3)

b. (3, 0)

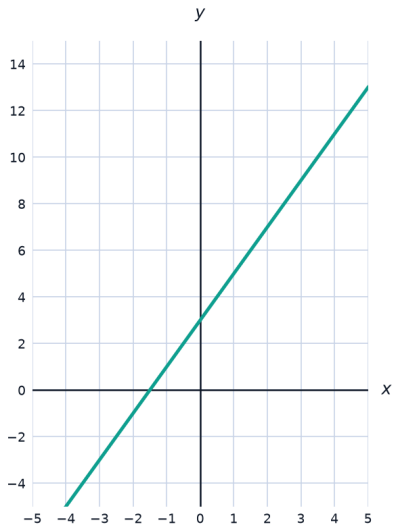
c. (1, 0)

d. (0, -3)

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41 [Challenging] • Target: MCQ — Substitution into formulas • Cross-Lesson: 6.01 Sequences & Patterns

[redacted] $f(x) = 2x + 3$[redacted] [redacted] [redacted] [redacted] $f(5)$[redacted] For the function $f(x) = 2x + 3$, what is the value of $f(5)$?



a. 8

b. 10

c. 13

d. 23

42 [Challenging] • Target: MCQ — Translating words into equations • Cross-Lesson: 7.01 Data Analysis & Probability

[redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted]. [redacted] [redacted] [redacted] [redacted] 36 [redacted] [redacted] 2 [redacted]. [redacted] [redacted] [redacted] [redacted] 48 [redacted] [redacted] 2 [redacted]. [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] 25 [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted]

43 [Challenging] • Target: MCQ — Word problems • Cross-Lesson: 8.01 Word Problems & Reasoning

[illegible] A contracting company has two major contracts. Contract 1 requires producing 1200 units in 30 days at a constant rate. Contract 2 requires producing 1800 units in 45 days at a constant rate. If the company works on both contracts simultaneously, starting on the same day, what is the minimum number of days the company can complete both contracts, assuming the company's production rate for each contract is constant and independent of the other, and the company can allocate its resources to work efficiently on both projects?

a. 30 [illegible]30 days

b. 45 [illegible]45 days

c. 60 [illegible]60 days

d. 75 [illegible]75 days

44 [Challenging] • Target: Evaluating expressions • Cross-Lesson: 1.01 Arithmetic & Number Sense

[illegible] $ax^2 + bx + c = 0$ [illegible] $x = 2$ [illegible] $x = 3$ [illegible] [illegible] $\frac{a+b+c}{a}$[illegible] If $ax^2 + bx + c = 0$ represents a quadratic equation, and $x = 2$ and $x = 3$ are its roots, what is the value of $\frac{a+b+c}{a}$?

a. 1

b. 5

c. 6

d. 1/6

45 [Challenging] • Target: Solving linear equations • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

[illegible] $f(x) = ax^2 + bx + c$. [illegible] $f(1) = 5$[illegible] $f(2) = 11$[illegible] [illegible] $f(3) = 19$[illegible] [illegible] $a + b + c$[illegible] [illegible] a[illegible] b[illegible] [illegible] c [illegible] Let the function be $f(x) = ax^2 + bx + c$. If $f(1) = 5$, $f(2) = 11$, and $f(3) = 19$, what is the value of $a + b + c$, given that a, b, and c are real constants?

a. 5

b. 7

c. 10

d. 12

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